
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

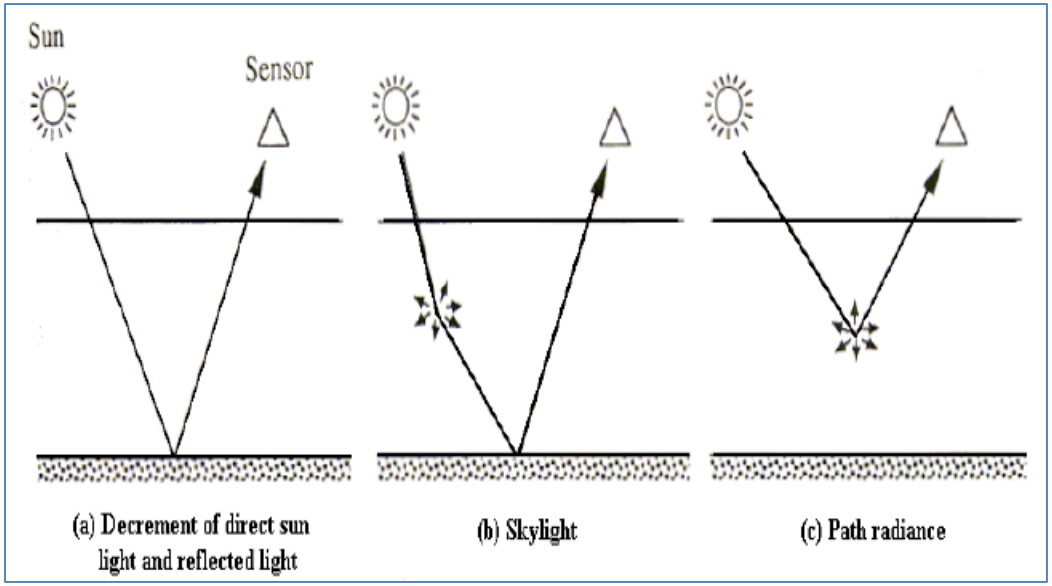
Radiometric / Atmospheric Distortions and Corrections

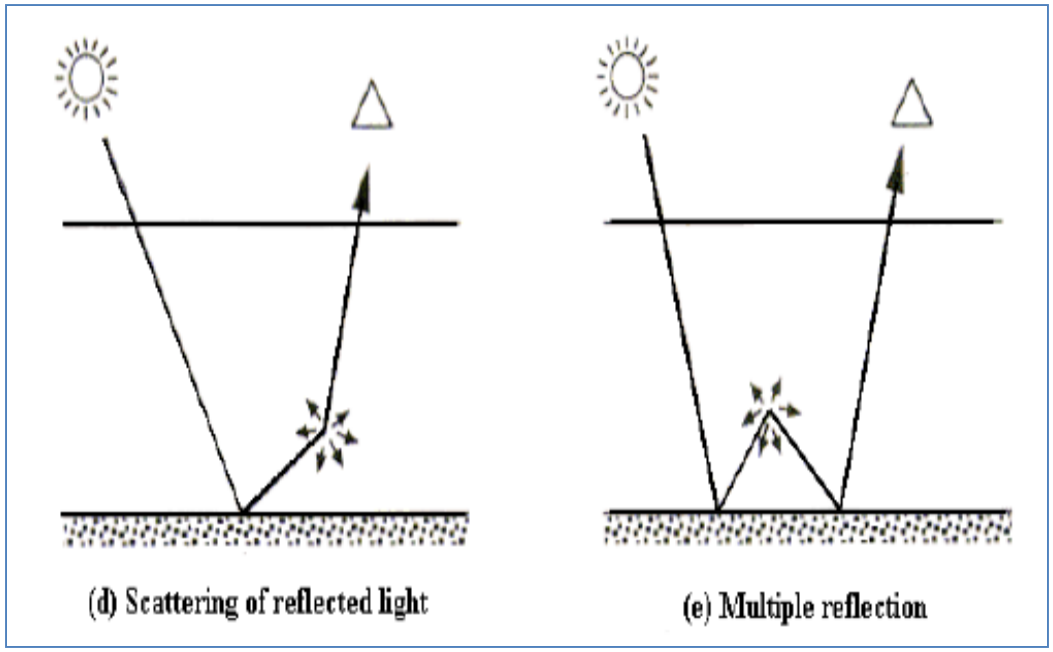
Radiometric Distortion

- The electromagnetic radiance arriving at the sensor goes through complex interactions
- **The solar radiance reaching the upper atmosphere depends on the**:
  - Longitude and latitude
  - Time (date and hour) of the year
- **Going through the atmosphere** (up and down), the Electromagnetic radiance is affected by processes of
  - Scattering
  - Absorption
- **The amount of incident radiance upon the surface**, depends on the:
  - Optical depth (height of the surface)
  - Slope and aspect of the surface

Atmospheric Distortion

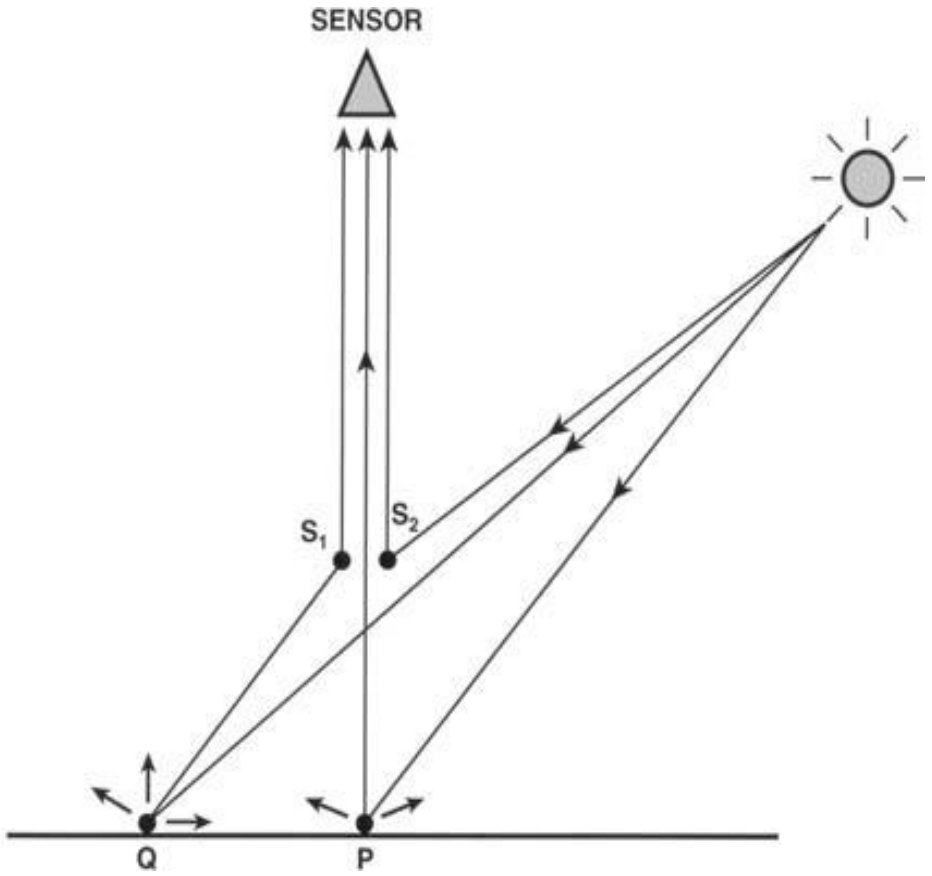
- **Ideally, radiant flux recorded by sensor in various bands is an accurate representation of the radiant flux leaving the features of interest, BUT**
  - Actually, it is not the case
- **Emitted or reflected EMR observed by a sensor on board an aircraft or spacecraft does not coincide with the energy emitted or reflected from the same object observed**
- **Sources**
  - **The atmosphere as a transmission medium**
  - ***Adjacency effect /adjacent pixels, the signal from a certain pixel partly mixed with signals from its surrounding pixels***

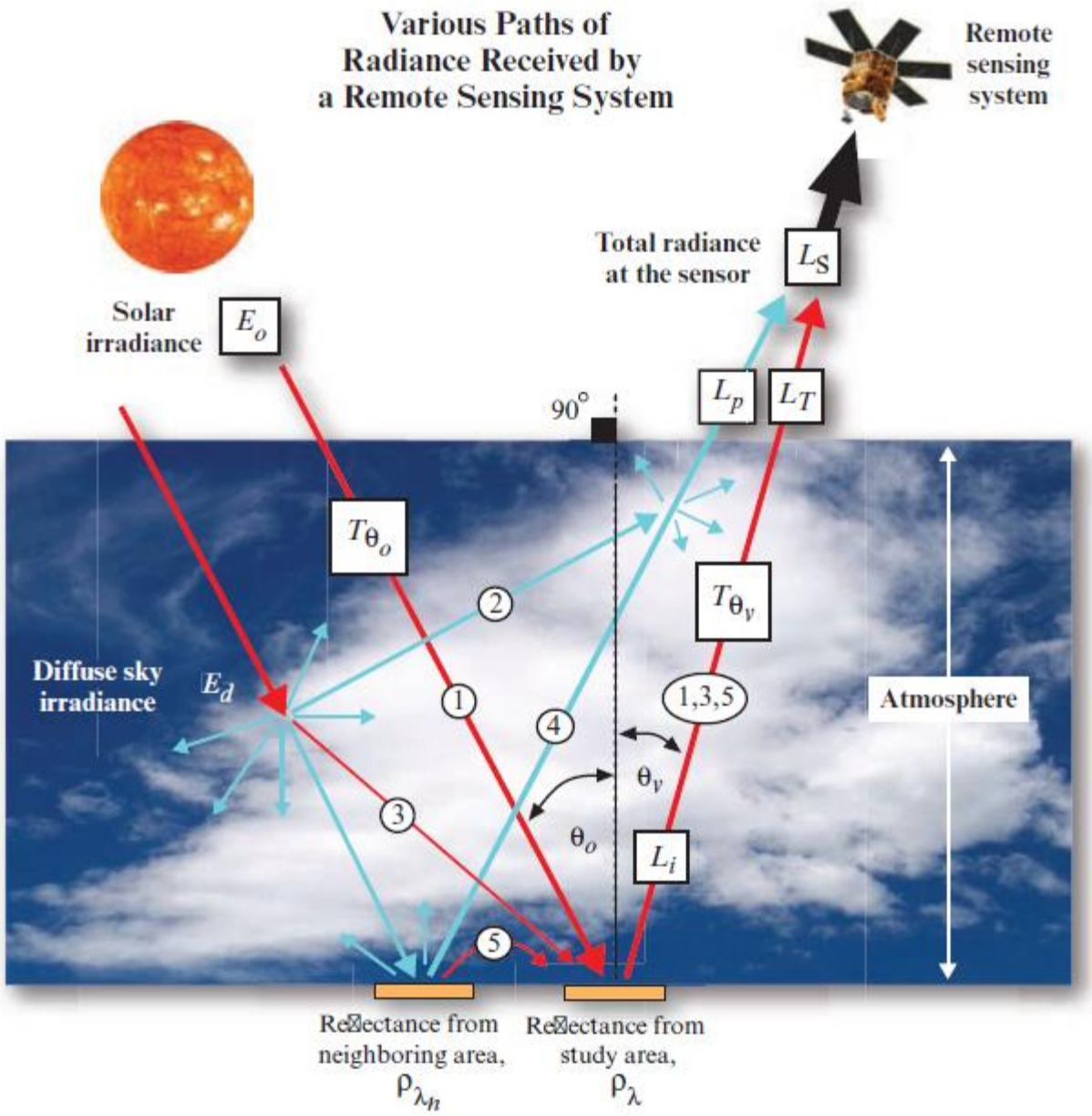

Sun
Sensor
(a) Decrement of direct sun light and reflected light
(b) Skylight
(c) Path radiance


(d) Scattering of reflected light
(e) Multiple reflection

Radiometric- Atmpspheric Distortion

- **A value recorded at given pixel location on a RS image is not a record of the true ground leaving radiance at that point**
- **Atmospheric path radiance**: Redirected EM energy within atmosphere to the FOV of sensor (S2)
- **Adjacency Effects**: Energy reflected from some other object(Q) but seen at sensor from a specific object (P)
- All the response appear to be from point P





Radiance (L_T) from paths 1, 3, and 5 contains intrinsic valuable spectral information about the target of interest. Conversely, the *path radiance* (L_p) from paths 2 and 4 includes diffuse sky irradiance or radiance from neighboring areas on the ground. This path radiance generally introduces unwanted radiometric noise in the remotely sensed data and complicates the image interpretation process.

Radiometric- Atmospheric Corrections

Atmospheric Corrections

- **The effect of atmospheric scattering lift the overall brightness of an image in each band**
- **Scattered path radiance introduces haze in the imagery and reduces contrast**
- **Atmospheric correction removes such haze and image intensity is improved**
- **Such Correction is also known as “Haze Removal”**
- **Haze compensation is designed to minimize the influence of path radiance effects**

Atmospheric Corrections

- **Atmospheric correction might be a necessary pre-processing technique in three cases**
  - **To compute a ratio of the pixel values in two bands**
  - **To relate upwelling radiance from a surface to some property of that surface. To do this the atmospheric component present in the signal recorded by the sensor must be estimated and removed**
  - **To compare results achieved from two or more different time images. Since the state of the atmosphere will undoubtedly vary , it is necessary to correct the radiance values recorded by the sensor for the effects of the atmosphere**

Atmospheric Corrections

- **Assumptions**
  - **Each band of the data for a given scene should have contained some pixels at or close to zero brightness value**
  - **Atmospheric effects and especially path radiance has added a constant values to each pixel**

Atmospheric Corrections

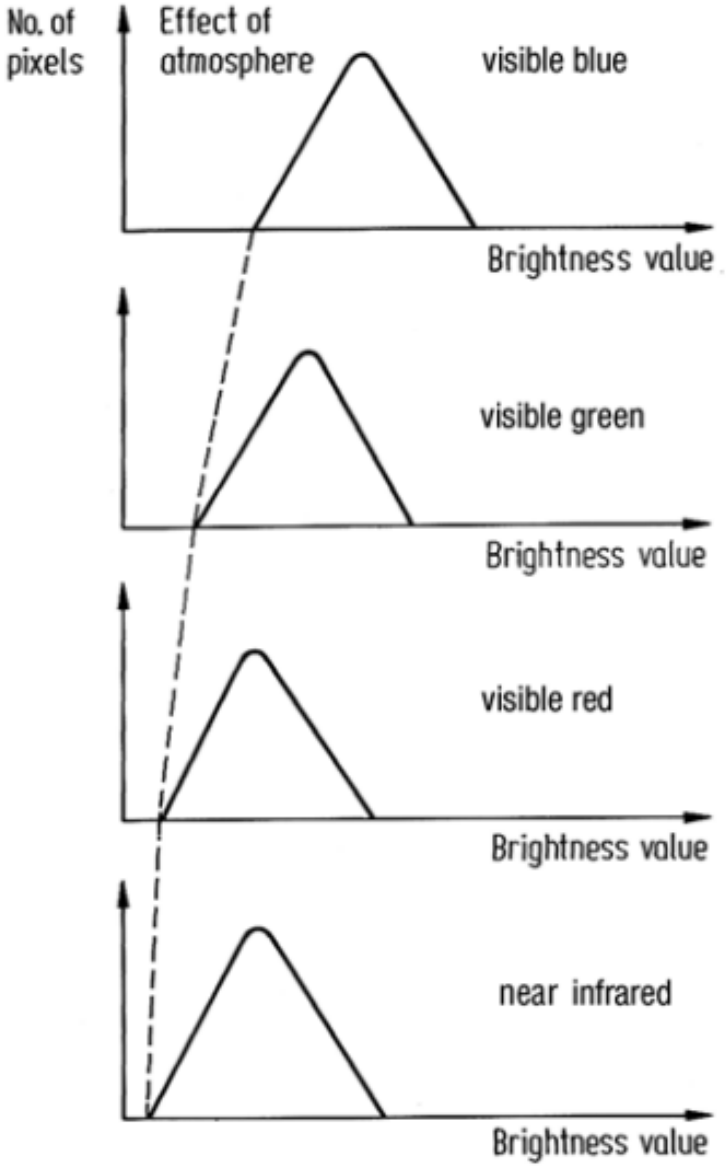
Image Based Method

- **Estimation of path radiance and its subtraction from the signal received by sensor**
  - **Histogram Minimum method (HMM) Or Dark object subtraction (DOS)**
  - **Regression Method**

Atmospheric Corrections

Histogram Minimum Method

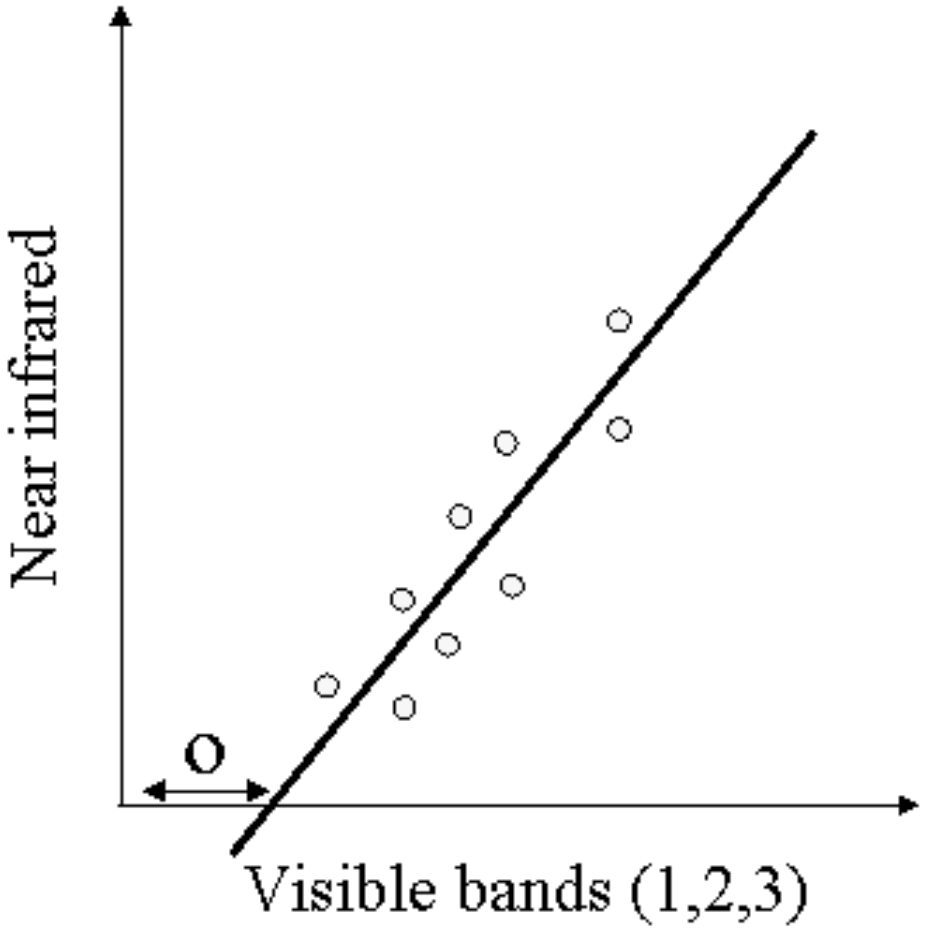
- **Histograms of pixel values in all bands are computed (should be of some areas of low reflectance (clear/ deep water)**
- **These pixels will have values very close to zero in near-infrared bands, and should have near-zero values in the other bands**
- **Histograms of these other bands will generally be seen to offset progressively towards higher levels**
- **These histograms of visible and near-infrared bands will help identify an approximation to the atmospheric path radiance in those bands**
- **These minimum values are subtracted from the respective image bands**



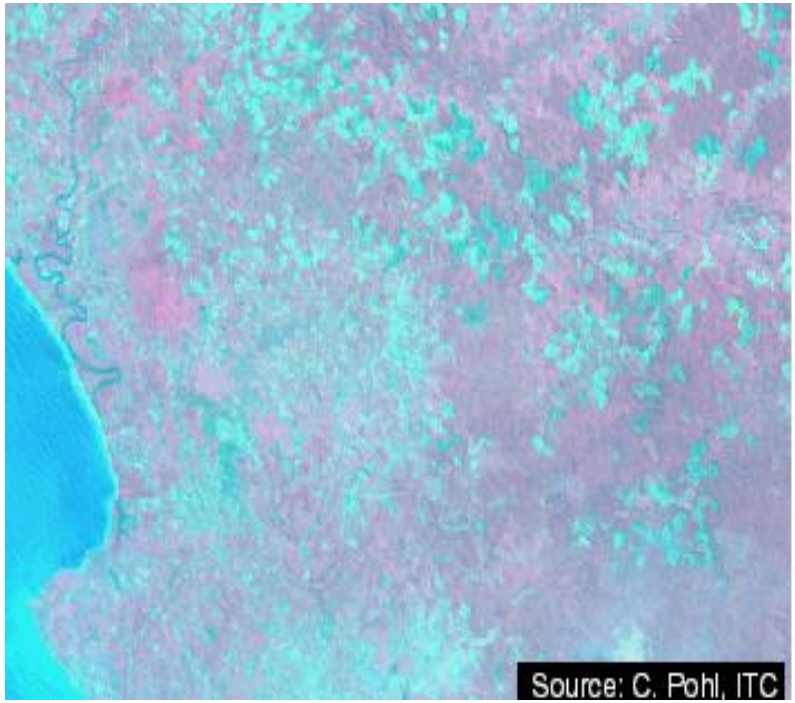
Effect of path radiance, resulting from atmospheric scattering, on the histograms of four bands of image data at different wavelengths

Regression Method

- Is applicable to dark pixel areas in the image
- Pixel values in NIR are plotted against the values in other bands
- Best Fit) is computed for each band using regression method
- An offset on x-axis for each regression represents an estimate of the atmosphere path radiance for the associated spectral band
- This intercept is taken as correction value for the specific band
- **Subtract offset from each visible band**
- Can be applied to local areas



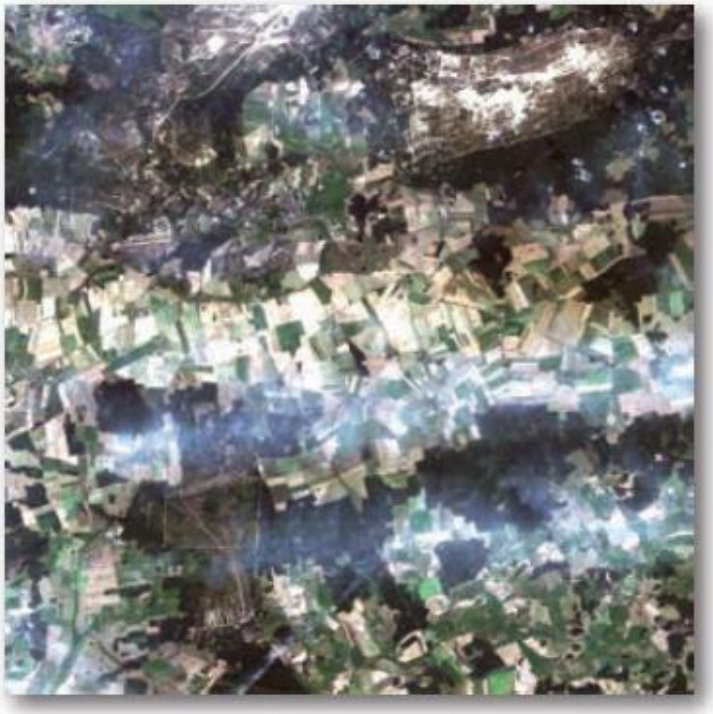
Haze Removal



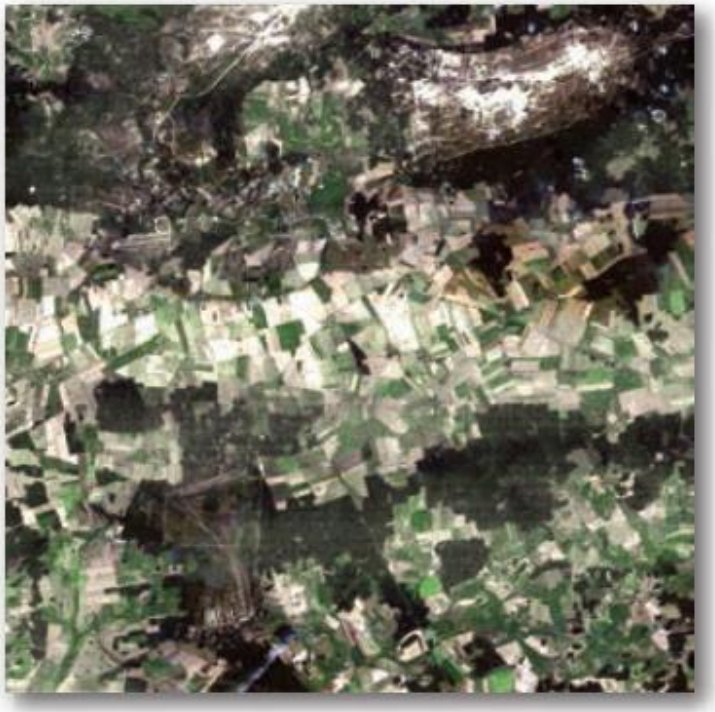
Before



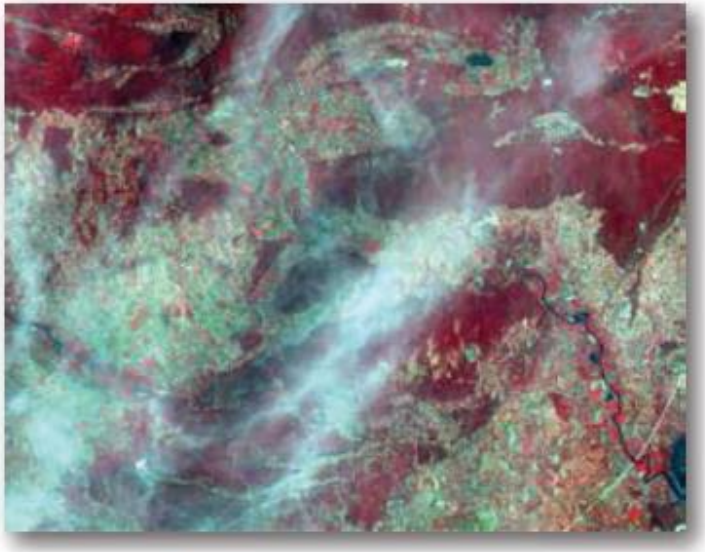
After



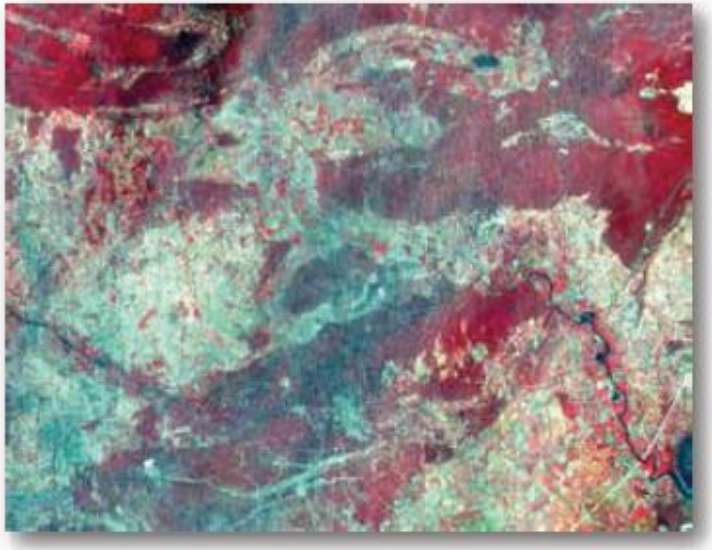
a. Before atmospheric correction.



b. After atmospheric correction.



c. Before atmospheric correction.



d. After atmospheric correction.

Haze Removal- Aerial Image



Before haze removal



After haze removal

Empirical Line Method

- **The Empirical Line calibration technique is used to force image data to match selected field reflectance**
- **This method requires ground measurements and/or knowledge (obtained at approximately the same time and on the same date as the remote sensing)**
- **Two or more ground targets, at least one light and one dark area are identified and reflectance is measured in the field**
- **The same two targets are identified in the image and average spectra are extracted for Regions of Interest**

In Situ Radiometric Data Collection





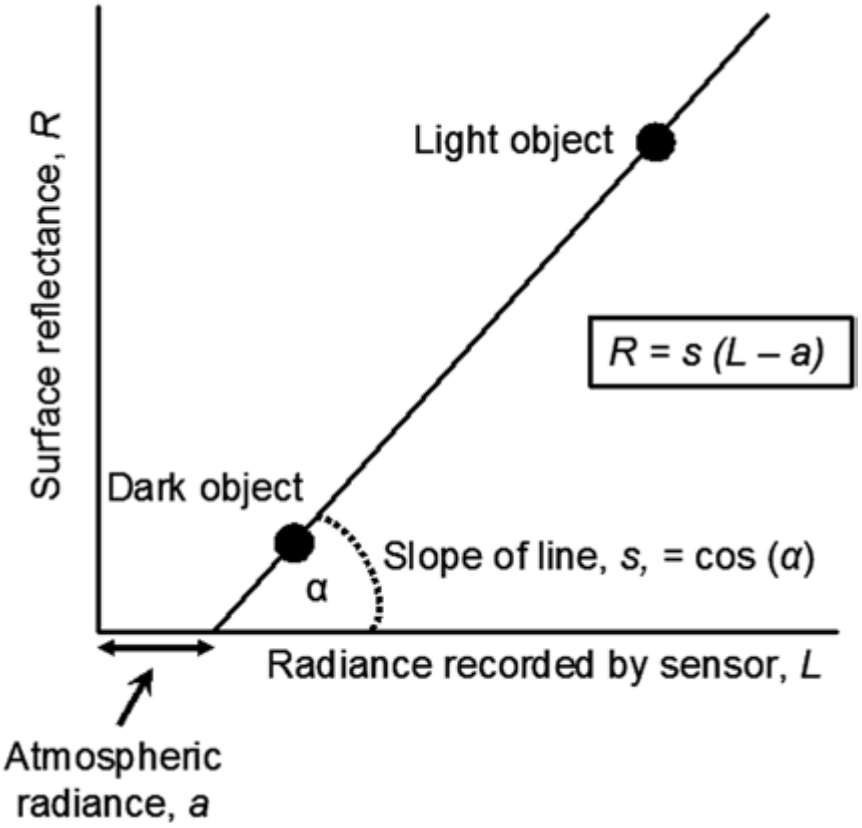
- Field crew taking a spectroradiometer measurement from a calibrated reflectance standard on the tripod
- 8×8 m black and white calibration targets

Empirical Line Method

- **A linear regression is calculated between the field reflectance spectra and the image radiance spectra (for each band separately) to determine a linear transform from radiance to reflectance for each band of the data set**
- **Gains and offsets calculated in the regression are applied to the radiance spectra for each pixel to produce apparent reflectance on a pixel-by-pixel basis**

Empirical Line Method

- Select dark and bright areas of the image
- Acquire ground-based reflectance measurements via a field spectrometer– If not available, spectral library values may be possible
- Fit regression
- Compute sensor radiance from image data, band by band for each pixel
- R=s(L-a)
- s=cos (α)
- L= senor radiance
- a= intercept of line joining bright and dark targets (atm radiance)



Atmospheric Corrections

- **If multi-temporal image data are being processed then they must be corrected for atmospheric effects to ensure that they are comparable**

Radiometric Calibration

Radiometric Calibration

- **Pixel values in commercially available satellite imagery represent the radiance of the surface in the form of Digital Numbers (DN) which are calibrated to fit a certain range of values**
- **Although DNs have practical value, they do not present brightness in the physical units**
- **When image brightness are expressed as DNs, each image becomes an individual entity without a defined relationship to other images or to features on the ground**

Radiometric Calibration

- **Essential in applications where measurement of absolute radiances is required**
- **Necessary when changes in the absolute reflectance of objects are to be measured over time using different sensors**
- **Conversion of DN to absolute radiance values is a necessary procedure for comparative analysis of several images taken by different sensors**
- **Since each sensor has its own calibration parameters used in recording the DN values, the same DN values in two images taken by two different sensors may represent two different radiance values or vice versa**

Conversion from Radiance to DN and back

- **Three parameters measured by a detector over the spectral bandwidth of the band**
  - **Range of DN values in the image**
  - **Lowest radiance (L_{min})**
  - **Highest radiances (L_{max})**
- **Most commonly, the data are distributed in 8-bit format corresponding to 256 DN levels**
- **L_{min}: is the spectral radiance corresponding to the minimum DN value (usually, a value of 0)**
- **L_{max}: is the radiance corresponding to the maximum DN (usually, the value of 255)**

Conversion from Radiance to DN

- **For each spectral band, the grey level or DNs, is related to the radiance (L) by the following equation:**

$$DN = (L_\lambda \times Gain) + D$$

  - **L: Measured Spectral radiance**
  - **G: Slope of the response function (Gain)**
  - **D: Intercept of the response function (offset, bias)**
  - **All the quantities in the equation are a function of wavelength**

- **Above information for each image is provided in the metadata files**

Conversion from DN to Radiance

- **The image DN can be converted to the physical quantity radiance (L, expressed in Wm2 sr-1) by equation:**

$$L_\lambda = (L_{max}\text{-}L_{min}/255) \times DN + L_{min}$$

Reflectance $$\rho_\lambda = \pi d^2 L\lambda / ESUN_\lambda \theta_E$$

- Where,
  - d: Earth-sun distance in astronomical units (147.76) million Km
  - $ESUN_\lambda$: mean solar irradiance for each spectral band
  - θ_E : sun elevation angle

Conversion from DN to Radiance in Landsat 8

- DN values can be converted to spectral radiance using the radiance scaling factors provided in the meta data file:
  - $L_\lambda = M_L * Q_{cal} + A_L$
- L_λ=Spectral radiance (W/(m2 * sr* μm))
- M_L=Radiance multiplicative scaling factor for the band, (RADIANCE_MULT_BAND_n from the metadata)
- A_L=Radiance additive scaling factor for the band, (RADIANCE_ADD_BAND_n from the metadata)
- Q_{cal}= pixel value in DN

Atmospheric Corrections

- **Mainly depends upon data applications**
- **Required if**
  - **Need calibrated analysis**
  - **Need to compare calibrated spectra across location and time**
  - **Need to compare calibrated spectra to spectral libraries**
- **May not if:**
  - **Only visual analysis is required**
  - **Only basic classification is required (obvious land covers)**
  - **Single image analysis**

Conclusion-Pre-Processing Tips

- **Before starting a project always:**
  - Derive as much information from the vendor on what has been done to the image supplied
  - Check carefully the header and all files supplied with the image
  - Visually check the image for visual atmospheric effects e.g., visible haze
- **Evaluate your needs:**
  - If not performing change detection, you do not need to correct for atmospheric effects
  - If producing an image for visual overlay with map data, then select bicubic re-sampling but never for quantitative analysis
- **Know your algorithms:**
  - Be aware of the implications, assumptions and limitations of a particular approach - for example, what a particular approach is correcting

References

- Introductory Digital Image Processing (Jensen): Chapter 6, Geometric correction of RS data: PP 125-135
- Computer Processing of RS Images (P M. Mather-1987): Chapter 4, Pre-processing of RS data: PP 125-142, sec 4.2.1
- Computer Processing of RS Images (P M. Mather-2005): Chapter 4, Geometric correction and registration: PP 87-107
- RS Digital Image Analysis, An Introduction (J A.Richrds-2nd): Chapter 2,Sources of Geometric distortion: pp 48-71

